EM-TCCM Course

Multiscale, Machine learning and QSAR Methods applied to biomolecules

25.-29.11.2024

Université Toulouse III - Paul Sabatier

Venue: Aspet, France

Course Coordination: Andrea Lombardi, Noelia Faginas Lago, Dipartimento di Chimica, Biologia e Biotecnologie, Università di Perugia, Perugia, Italy

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Local organizers: Prof. Stefano Evangelisti, Nicolas Suaud, IRSAMC, Université Toulouse III - Paul Sabatier, Toulouse, France

Pedagogical support: Peter Reinhardt, Lab. Chimie Théorique, Sorbonne Université, Paris, France

With the support of the
Erasmus+ Programme
of the European Union

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Venue:

Le Bois Perché (https://www.boisperche.com)
Las Vignes 31160 Aspet, France

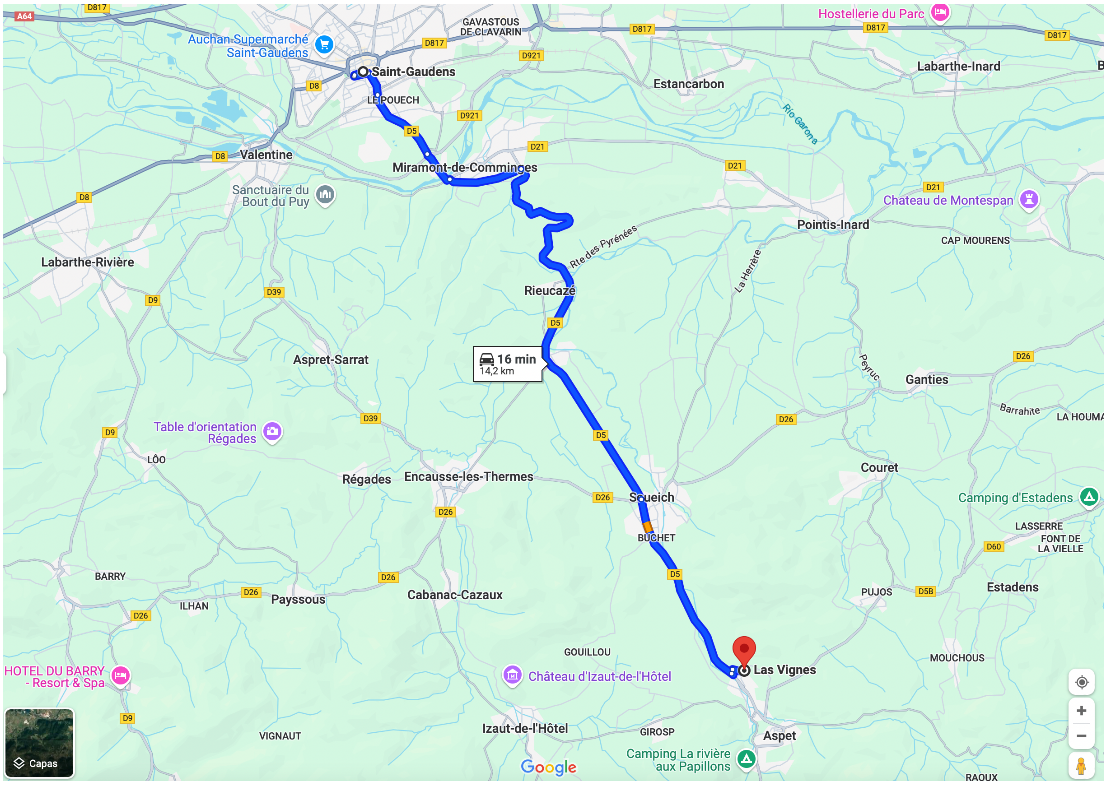
How to get there?

If you fly to Toulouse-Blagnac:

1. Get a transportation (bus/taxi) to the train station (Toulouse Matabiau) (ca. 30 min)
2. From Toulouse Matabiau get a train to Saint-Gaudens (Intercites/TER) (ca. 1h): Note that the last train from Toulouse Matabiau to St Gaudens is at 20h30 and the first train from St Gaudens to Toulouse Matabiau is at 07h05
3. From Saint-Gaudens to Aspet and viceversa: We'll pick you up with a car and/or we'll arrange taxis.

If you arrive Toulouse Matabiau: See steps 2) and 3) above

Once at Saint-Gaudens, you will be picked up at the train station by arranged taxis:



Meals:

During your stay at "Le bois perché" all meals (breakfast/lunch/dinner) will be covered.

Summary of the activities

The course will deliver the basics of Machine Learning and more advanced applications, by theoretical lectures in the morning, followed by corresponding hands-on sessions in the afternoon. An additional course dedicated to Reaction Dynamics will be held online, scheduled on November 13-15 and December 6, 2024. See below for the corresponding Zoom links

List of the lectures

Introduction to Machine and Deep Learning using python, Loriano Storchi, Università G. D'Annunzio, Italy

Structure-Based Design and QSAR using GRID Molecular Interaction Fields, Simon Cross, Molecular Horizon srl, Italy

Neural networks potentials for reactive molecular simulations, Damien Laage, CNRS, UMR PASTEUR, Paris, France

Machine Learning in Computational Chemistry: A Python and Scikit-learn introduction, Marco Pezzella, Dipartimento di Fisica e Geologia, Università di Perugia, Perugia, Italy

QMMM simulations to explore enzymatic reactions, Marie Brut, LAAS-CNRS, Toulouse, France

Introduction to Kinetic Monte Carlo and Exploration of Potential Energy Surface, Anne Hémeryck, LAAS-CNRS, Toulouse, France

QSAR and 3D QSAR, Rino Ragno, Rome University "La Sapienza", Rome, Italy

High-Performance MD, Ferran Feixas, Institut de Quimica Computacional i Catàlisi, University of Girona, Spain

Multiscale Simulations of Biomolecular Systems. From Atoms to Blobs and Back, Ioana ILIE, Van 't Hoff Institute for Molecular Sciences University of Amsterdam, Netherlands

Artificial Intelligence & Machine Learning in chemistry - Part I, Romuald Poteau, LPCNO, Toulouse, France

Online courses

Course: Chemical reaction dynamics: from fundamentals to new insights

Date: 13 - 14 - 15 November 2024 (Time 11 - 13)
https://us06web.zoom.us/j/83640233901?pwd=s4vd8VKGlqHs5RMmWOqNMEbcnYgZW5.1
ID: 836 4023 3901
Passw: 186504

Date: 6 December 2024 (Time 15-17)
https://us06web.zoom.us/j/83434841095?pwd=056pgtqlB5eY0mmauDhx8b3q6WipjX.1
ID: 834 3484 1095
Passw: 730338

School:

Multi-scaling, Machine learning and QSAR - Course
25 to 29 nov 2024 08:00 a. m.
https://us06web.zoom.us/j/85650233984?pwd=SJ1hkGdEVBnlCu3krnWOIo9dOqXPI3.1
ID: 856 5023 3984
Passw: 183314

Program at glance

	Multiscale, Machine Learning and QSAR Methods Applied to Biomolecules 25-29 November, Aspet (France)				
Time	Monday 25	Tuesday 26	Wednesday 27	Thursday 28	Friday 29
08:00-09:45	"Introduction to Machine and Deep Learning using python" Loriano Storchi Universita' G. D'Annunzio (Italy)	"Neural networks potentials for reactive molecular simulations" Damien Laage CNRS, UMR PASTEUR (Paris)	QMMM simulations to explore enzymatic reactions Marie Brut LAAS-CNRS (Toulouse)	"QSAR and 3D QSAR" Ragno Rino Sapienza Rome University (Italy)	"Multiscale Simulations of Biomolecular Systems. From atoms to Blobs and Back" Ioana Ilie Van 't Hoff Institute for Molecular Sciences University of Amsterdam (Netherlands)
09:45-10:15	**BREAK**				
10:15-12:00	"Structure-Based Design and QSAR using GRID Molecular Interaction Fields" Simon Cross Molecular Discovery (Italy)	"Machine Learning in Computational Chemistry: A Python and Scikit-learn introduction" Marco Pezzella Dipartimento di Fisica e Geologia - FISGEO (Italy)	"Introduction to Kinetic Monte Carlo and Exploration of Potential Energy Surface" Anne Hémeryck LAAS-CNRS (Toulouse)	"High-Performance MD" Ferran Feixas Institut de Química Computacional i Catàlisi University of Girona (Spain)	Artificial Intelligence & Machine Learning in chemistry - Part I Romuald Poteau LPCNO, Toulouse, France
12:00-14:00	**LUNCH**				
14:30-16:30	"Introduction to Machine and Deep Learning using python" **Hands-on** Loriano Storchi Universita' G. D'Annunzio (Italy)	Neural networks potentials for reactive molecular simulations **Hands-on** Damien Laage CNRS, UMR PASTEUR (Paris)	"QMMM Simulations to Explore Enzymatic Reactions" **Hands on** Marie Brut LAAS-CNRS (Toulouse)	"QSAR and 3D QSAR" **Hands-on** Ragno Rino Sapienza Rome University (Italy)	"Multiscale Simulations of Biomolecular Systems. From Atoms to Blobs and Back " **Hands-on** Ioana Ilie Van't Hoff Institute for Molecular Sciences University of Amsterdam(Netherlands)
16:30-17:00			**BREAK**		
17:00-19:00	"Structure-Based Design and QSAR using GRID Molecular Interaction Fields" **Hands-on** Simon Cross Molecular Discovery (Italy)	"Machine Learning in Computational Chemistry: A Python and Scikit-learn introduction" **Hands-on** Marco Pezzella Dipartimento di Fisica e Geologia (Italy)	"Exploration of Potential Energy Surface using Activation Relaxation Technique" **Hands on** Anne Hémeryck LAAS-CNRS (Toulouse)	"High-Performance MD" **Hands on** Ferran Feixas Institut de Química Computacional i Catàlisi University of Girona (Spain)	Artificial Intelligence & Machine Learning in chemistry - Part II **Hands-on** Romuald Poteau LPCNO, Toulouse, France

Abstracts of the lectures

Introduction to Machine and Deep Learning using python

Loriano Storchi [1]
[1] *University G. D'Annunzio of Chieti-Pescara, via dei Vestini 31, 66100 Chieti*
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Machine learning (ML) is a rapidly evolving field that has the potential to revolutionize many aspects of our lives. ML techniques allow computers to learn from data without being explicitly programmed. This makes them ideal for solving problems that are too complex or time-consuming for humans to solve manually.

This presentation will introduce the fundamental concepts of ML and provide an overview of some of the most common ML techniques. We will discuss supervised learning, unsupervised learning, and reinforcement learning. with a specific focus on Supervised ML covering both the most important Deep and non-Deep Learning approaches We will also explore some working examples [1-5].

By the end of this presentation, you will have a basic understanding of ML and how it can be used to solve real-world problems. The whole lecture will focus on the coding part with several code snippets presented and explained.

References

[1] Loriano Storchi, Gabriele Cruciani, Simon Cross, "DeepGRID: Deep Learning using GRID descriptors for BBB prediction", Journal of Chemical Information and Modeling, DOI: 10.1021/acs.jcim.3c00768 (2023)
[2] Tommaso Tedeschi, Marco Baioletti, Diego Ciangottini, Valentina Poggioni, Daniele Spiga, Loriano Storchi, Mirco Tracolli, "Smart Caching in a Data Lake for High Energy Physics Analysis", Journal of Grid Computing, DOI: 10.1007/s10723-023-09664-z (2023)
[3] Qizhen Hong, Loriano Storchi, Massimiliano Bartolomei, Fernando Pirani, Quanhua Sun, Cecilia Coletti, "Inelastic N2+H2 collisions and quantum-classical rate coefficients: large datasets and machine learning predictions" The European Physical Journal D, DOI: 10.1140/epjd/s10053-023-00688-4 (2023)
[4] Udaykumar Gajera, Loriano Storchi, Danila Amoroso, Francesco Delodovici, Silvia Picozzi "Towards machine learning for microscopic mechanisms:a formula search for crystal structure stability based on atomic properties" Journal of Applied Physics, DOI: 10.1063/5.0088177 (2022)
[5] Sara Tortorella, Emanuele Carosati, Giovanni Bocci, Simon Cross, Gabriele Cruciani, Loriano Storchi, "Combining Machine Learning and Quantum Mechanics Yields More Chemically-Aware Molecular Descriptors for Medicinal Chemistry Applications", Journal of Computational Chemistry, DOI: 10.1002/jcc.26737 (2021)

Structure-Based Design and QSAR using GRID Molecular Interaction Fields

Simon Cross [1]
[1] *Molecular Discovery, UK*
simon@moldiscovery.com

Students will learn how to use the GRID 2021 software[1] to explore protein binding sites for energetically favourable hotspots which can be used for ligand design, including the automated FragExplorer approach[2] for exploring novel R-Groups that match the GRID hotspots. Students will also learn how to use FLAP[3] to build 3D-QSAR models for a series of active ligands, using a machine-learning approach to identify the ligand MIFs that contribute in favourable and unfavourable ways to ligand inhibition. Students will also learn how to rationalise the results of this in the context of the receptor using WaterFLAP network prediction.

References

[1] Goodford P. J. A computational procedure for determining energetically favorable binding sites on biological important macromolecules. J. Med. Chem. 1985, 28, 849-857.
[2] Cross S. and Cruciani G. FragExplorer: GRID-Based Fragment Growing and Replacement. J. Chem. Inf. Model. 2022, 62, 5, 1224–1235. https://doi.org/10.1021/acs.jcim.1c00821
[3] Baroni M.; Cruciani G.; Sciabola S.; Perruccio F.; Mason J. S. A Common Reference Framework for Analyzing/Comparing Proteins and Ligands. Fingerprints for Ligands And Proteins (FLAP): Theory and Application. J. Chem. Inf. Model. 2007, 47, 279-294.

Neural Network Potentials for Reactive Molecular Dynamics

Damien Laage [1]
[1] *PASTEUR, Department of Chemistry, Ecole normale supérieure, PSL University, Sorbonne University, CNRS, Paris (France)*
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In this course we will show how recent developments in machine learning and deep neural networks can be used to perform reactive molecular dynamics simulations at a high electronic structure level but for a fraction of the computational cost [1-4]. We will first introduce the challenges faced by simulations of chemical reactions in condensed phases. We will then provide an introduction to deep neural network potentials, their structure, the definition of descriptors, their training, the construction of the training set in the case of reactive systems, and their combination with molecular dynamics simulations. We will then show how these methods can be extended to describe electrostatics and to calculate vibrational spectra. We will then give an overview of typical applications of these methods. The outline of the class will be:

1. Challenges of simulating chemical reactions in condensed phases
2. Introduction to deep neural network potentials: structure, descriptors, training
3. Extension to electrostatics and spectroscopy
4. Typical applications

The practical part of the course will be devoted to the training of a neural network potential to describe the dissociation of an acid at the air-water interface[5] using DeePMD[6] and LAMMPS[7].

References

[1] Machine learning for molecular simulation F. Noé, A. Tkatchenko, K.-R. Müller, C. Clementi *Annu. Rev. Phys. Chem.* **2020** *71*, 361-390
[2] Machine learning force fields O.T. Unke, S. Chmiela, H.E. Sauceda, M. Gastegger, I. Poltavsky, K.T. Schütt, A. Tkatchenko, K.-R. Müller *Chem. Rev.* **2021** *121*, 10142-10186
[3] Neural network potentials: a concise overview of methods E. Kocer, T W Ko, J. Behler *Annu. Rev. Phys. Chem.* **2022** *73*, 163-186
[4] Neural network potentials for chemistry: concepts, applications and prospects S. Käser, L.I. Vazquez-Salazar, M. Meuwly, K. Töpfer *Digital Discovery* **2023** *2*, 28-58
[5] Acids at the edge: why nitric and formic acid dissociations at air-water interfaces depend on depth and on interface specific area M. de la Puente, R. David, A. Gomez, D. Laage *J. Am. Chem. Soc.* **2022** *144*, 10524-10529
[6] DeePMD-kit: a deep learning package for many-body potential energy representation and molecular dynamics H. Wang, L. Zhang, J. Han, W. E *Comp. Phys. Commun.* **2018** *228*, 178-184
[7] Fast parallel algorithms for short-range molecular dynamics S. Plimpton *J. Comp. Phys.* **1995** *117*, 1-19

Machine Learning in Computational Chemistry: A Python and Scikit-learn introduction

Marco Pezzella [1]

[1] *Dipartimento di Fisica e Geologia, Università Di Perugia, Via Alessandro Pascoli SNC, 06123 Perugia, Italy*
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Machine learning is becoming increasingly important in every aspect of our lives, from everyday applications (e.g. social media, job applications, search engines) to scientific research, as demonstrated by the Nobel Prizes in Physics[1] and Chemistry[2] in 2024. But how does it work? What lies behind these models?

In this lecture and the subsequent hands-on session, we will explore how to understand and build simple machine learning models, as well as the tools required to implement them. We will begin with an introduction to Python, discussing its pros and cons as a programming language, and how to utilise environments and libraries. Among the various libraries, Scikit-learn[3] will be a primary focus due to its ease-of-use together with other libraries.

Following some general application examples, we will focus mainly on applications in molecular spectroscopy. Spectroscopy models are particularly appealing, as they contain large amounts of data. We will use materials from NIST[4], MARVEL[5], and ExoMol[6] as a starting point for our work.

References

[1] "For foundational discoveries and inventions that enable machine learning with artificial neural networks", The Nobel Committee for Physics, https://www.nobelprize.org/uploads/2024/09/advanced-physicsprize2024.pdf.

[2] "Computational Protein Design and Protein Structure Prediction", The Nobel Committee for Chemistry, https://www.nobelprize.org/uploads/2024/10/advanced-chemistryprize2024.pdf.

[3] "Scikit-learn Machine Learning in Python", https://scikit-learn.org/stable/ (2024).

[4] "NIST Chemistry WebBook, NIST Standard Reference Database Number 69", U.S. Secretary of Commerce on behalf of the United States of America, https://doi.org/10.18434/T4D303, 2023.

[1] "MARVEL: measured active rotational–vibrational energy levels", T. Furtenbacher, A.G. Császár, J. Tennyson, *J. Mol. Spectrosc.* **245**, 115–125 (2007).

[6] "The 2024 release of the ExoMol database: molecular line lists for exoplanet and other hot atmospheres", J. Tennyson *& al.*, *J. Quant. Spectrosc. Radiat. Transf.*, **326**, 109083 (2024).

Hybrid QM/MM simulations to explore enzymatic reactions

Marie Brut [1]

[1] [1]*LAAS-CNRS, 7 avenue du Colonel Roche, 31031 Toulouse Cedex 4, France*

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Hybrid Quantum Mechanics / Molecular Mechanics simulations (QM/MM) allow to combines the accuracy of ab initio QM calculations and the efficiency of MM approaches, a requirement to study chemical reactions in large molecular like proteins.

In this course, we will first discuss the general QM/MM methodology with a special focus on its use in enzymatic reactions through various examples. Then, we will introduce the Amber software and work on practical examples to illustrate the different possibilities offered by this package, as well as certain limits and good practices.

References

[1] "QM/MM methods for biomolecular systems." H.M. Senn and W. Thiel. Angewandte Chemie International Edition 48.7 (2009): 1198-1229.
[2] "Biomolecular simulations: From dynamics and mechanisms to computational assays of biological activity." D.J. Huggins et al. Wiley Interdisciplinary Reviews: Computational Molecular Science 9.3 (2019): e1393.
[3] "Biomolecular QM/MM Simulations: What Are Some of the “Burning Issues”?." Q. Cui et al. The Journal of Physical Chemistry B 125.3 (2021): 689-702.
[4] "Harder, better, faster, stronger: large-scale QM and QM/MM for predictive modeling in enzymes and proteins." V. Vennelakanti, et al. Current Opinion in Structural Biology 72 (2022): 9-17.
[5] "Introduction to QM/MM simulations." G. Groenhof. Biomolecular Simulations (2013): 43-66.

Introduction to Kinetic Monte Carlo for Atomistic Simulations: Exploring the Potential Energy Surface with the Activation-Relaxation Technique

Anne Hemeryck
LAAS-CNRS, Université de Toulouse, CNRS, Toulouse, France

Kinetic Monte Carlo (KMC) is a numerical method of great importance for atomistic simulations, particularly useful for studying diffusion processes and chemical reactions at the atomic scale. Unlike molecular dynamics approaches, KMC allows the simulation of phenomena over much longer time scales by focusing on rare events and skipping less relevant intermediate dynamics. However, the effectiveness of this method heavily depends on the accuracy with which one can describe the potential energy surface (PES), representing the energy of all possible atomic configurations, even activation barriers. Constructing these surfaces is a major challenge: PESs are highly multidimensional, complex, and require detailed knowledge of interatomic interactions. The course will focus on the fundamentals of the KMC technique and how the use of the Activation-Relaxation Technique (ART) enables efficient exploration of the PES. Practical sessions will be dedicated to applying the ART method using both DFT software and Molecular Dynamics code.

References

Exploring potential energy surfaces to reach saddle points above convex regions - Miha Gunde, Antoine Jay, Matic Poberznik, Nicolas Salles, Nicolas Richard, Georges Landa, Normand Mousseau, Layla Martin Samos, Anne Hémeryck, *The Journal of Chemical Physics* **160** (2024) 232501 - doi: https://doi.org/10.1063/5.0210097

Ligand-based 3-D QSAR models: development and validation

Rino Ragno [1]
[1]*Department of Drug Chemistry and Technology, Sapienza University of Rome, Piazzale Aldo Moro 5 00185 Rome, Italy*
e-mail: rino.ragno@uniroma1.it

Three-dimensional quantitative structure-activity relationships (3-D QSAR) are computational techniques that aid in correlating the 3D structure of a series of molecule with their properties. In the field of medicinal chemistry, the property is mostly represented by a measured bioactivity (IC_{50} or K_i).
In the lecture the 3-D QSAR theoretical aspect will be elucidated with example and practical aspects.
In the practical section the student will be guided to develop a 3-D QSAR model using experimental data taken from a published article using the web portal www.3d-qsar.com [1,2].

For further details the students have to refer to the listed references.

References

[1] Ragno, R., Esposito, V., Di Mario, M., Masiello, S., Viscovo, M., and Cramer, R.D. Teaching and Learning Computational Drug Design: Student Investigations of 3D Quantitative Structure–Activity Relationships through Web Applications. Journal of Chemical Education 2020 97 (7), 1922-1930, DOI: 10.1021/acs.jchemed.0c00117.

[2] Ragno, R. www.3d-qsar.com: a web portal that brings 3-D QSAR to all electronic devices—the Py-CoMFA web application as tool to build models from pre-aligned datasets. J Comput Aided Mol Des 33, 855–864 (2019). https://doi.org/10.1007/s10822-019-00231-x.

Advanced Molecular Dynamics Simulations

Ferran Feixas[1]
[1]*Institut de Química Computacional i Catàlisi and Departament de Química, Universitat de Girona (Spain)*

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In this course, we will introduce the basics of advanced molecular dynamics simulation techniques to efficiently explore the conformational space of complex (bio)molecules and biochemical processes. First, we will introduce the need of enhanced sampling methods. Second, we will discuss the theoretical basis of different strategies based on hardware and software advances that have been developed to enhance conformational sampling and sample rare events.[1-3] Enhanced sampling methods can be divided into two groups: the ones based on the definition of set of collective variable or reaction coordinate, and the unconstrained methods. Then, we will learn how to run and extract relevant information from these simulations using machine learning based tools.[4] Finally, we are going to learn how these techniques can be used to reconstruct key steps of complex biochemical processes.[5] The outline of the class will be:

1- Introduction to the sampling problem and unbiased MD simulations
2- Basics of enhanced sampling methods: reaction coordinate-based and unconstrained techniques
3- Machine-learning based tools to run and analyze (un)biased MD simulations
4- Applications of enhanced sampling methods to the study of biochemical processes.

The practical part of the course will be focused on exploring the conformational space of a simple molecule using different enhanced sampling methods implemented in the Amber package. The protocols used in the practical session will be applicable to the study of other relevant (bio)chemical processes.

References

[1] Hénin, J.; Lelièvre, T.; Shirts, M. R.; Valsson, O.; Delemotte, L. Enhanced Sampling Methods for Molecular Dynamics Simulations. *Living Journal of Computational Molecular Science*, **2022**, *4*, 1583. https://doi.org/10.33011/livecoms.4.1.1583

[2] Kamenik, A.S.; Linker, S. M.; Riniker, S. Enhanced sampling without borders: on global biasing functions and how to reweight them. *Physical Chemistry Chemical Physics*, **2022**, 24, 1225-1236 https://doi.org/10.1039/D1CP04809K

[3] Miao, Y.; McCammon, J. A. Unconstrained enhanced sampling for free energy calculations of biomolecules: a review. *Molecular Simulations*, **2016**, 42, 1046-1055 https://doi.org/10.1080/08927022.2015.1121541

[4] Sittel, F.; Stock, G. Identification of collective variables and metastable states of protein dynamics. *J. Chem. Phys.*, **2018**, 149, 150901 https://doi.org/10.1063/1.5049637

[5] Calvó-Tusell, C.; Maria-Solano, M. A.; Osuna, S.; Feixas, F.Time Evolution of the Millisecond Allosteric Activation of Imidazole Glycerol Phosphate Synthase. *J. Am. Chem. Soc.* **2022**, 144, 7146-7159 https://doi.org/10.1021/jacs.1c12629

Multiscale Simulations of Biomolecular Systems. From Atoms to Blobs and Back

Ioana ILIE[1]
[1]Van 't Hoff Institute for Molecular Sciences
University of Amsterdam, Amsterdam, The Netherlands

Protein misfolding and aggregation are associated with the onset of neurodegenerative disorders such as Alzheimer's, Parkinson's and Creutzfel-Jakob's disease. To date no cure exists for neurodegenerative diseases and therapeutic interventions give limited symptomatic relief, rather than prevention. Different aggregates are associated with neurotoxicity. A better understanding of the physicochemical properties that govern the assembly mechanism of the early oligomeric species will aid in understanding their role in toxic propagation.

Here, I present our computational efforts to understand the aggregation mechanisms of polypeptides associated with neurodegenerative diseases from a multiscale perspective. First, we investigate the aggregation mechanisms of alpha-synuclein, the protein associated with Parkinson's disease. I will introduce a novel coarse-grained model for amyloidogenic polypeptides and use Brownian dynamics simulations to gain insight into the physical mechanisms of assembly into oligomeric species. Lastly, I present a novel iterative approach to design cyclic peptides that bind to soluble proteins with the aim of inhibiting the toxic propagation.

Artificial Intelligence and Machine Learning in Chemistry

Romuald Poteau [1]
[1] *LPCNO (INSA-UT3-CNRS, UMR 5215), INSA, 135 avenue de Rangueil, 31077 Toulouse Cedex, France*
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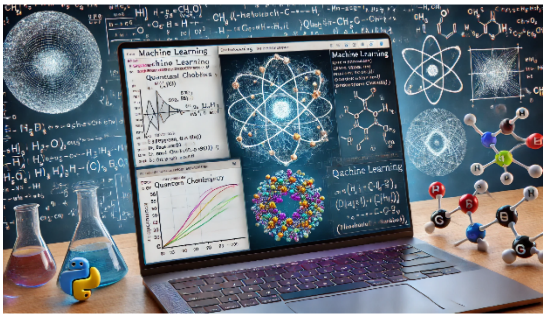
Machine learning (ML) techniques have been rapidly gaining considerable attraction in physical and chemical sciences. Promising – and sometimes impressive – applications are regularly published, and several review articles report on advances in this field.[1] In several cases, Artificial Intelligence (AI)/ML appears to be an accelerator for the design and optimization of new species or processes. ML approaches can also contribute to discovering patterns in data. A new paradigm seems to also arise in sciences of matter,[2] that aims to promote a more data-intensive and systematic research approach. This gives access to hybrid knowledge-driven and data-driven research approaches.



These "talktorials", a portmanteau for talks and tutorials, will explore the integration of some machine learning (ML) techniques into the field of chemistry. Through **Python-based tutorials embedded in Jupyter notebooks**, participants will gain hands-on experience with fundamental ML algorithms and their applications in chemistry, including predictive modeling and molecular property estimation. The session will cover key methodologies such **as regression, classification, and clustering**. In addition to supervised ML, these tutorials will also introduce **unsupervised ML techniques, focusing on auto-encoders**, still applied to physical chemistry properties.

All Jupyter notebooks are **freely available on the pyPhysChem github repository**.[3] **All participants will be required to run the notebooks on their own computers in parallel with the live tutorials**. More experienced participants will be able to experiment with the tuning of some hyperparameters. A basic knowledge of Python is a prerequisite for such talktorials, and participants must have a Python distribution installed on their computers. To assist with this, instructions and an introduction to Python will be provided through a series of video tutorials, accessible via a YouTube playlist.[4]

References

[1] (a) K. T. Butler, D. W. Davies, H. Cartwright, O. Isayev, A. Walsh, Machine learning for molecular and materials science, *Nature* 2018, **559**, 547-555 ; (b) J. G. Freeze, H. R. Kelly, V. S. Batista, Search for Catalysts by Inverse Design: Artificial Intelligence, Mountain Climbers, and Alchemists, *Chem. Rev*. 2019, **119**, 6595-6612 ; (c) P. Schlexer Lamoureux, K. T. Winther, J. A. Garrido Torres, V. Streibel, M. Zhao, M. Bajdich, F. Abild-Pedersen, T. Bligaard, Machine Learning for Computational Heterogeneous Catalysis, *ChemCatChem* 2019, **11**, 3581-3601 ; …………

[2] T. Hey, K. M. Tolle, S. Tansley, The Fourth Paradigm: Data-intensive Scientific Discovery, Microsoft Research 2009.

[3] N. Bernard, M. Charnay, S. Christodoulou, I. C. Gerber*, F. Jolibois* and R. Poteau*, *Python in the Physical Chemistry lab (pyPhysChem) github repository*, release v. 1.9.0 (2024), https://github.com/rpoteau/pyPhysChem

[4] https://youtube.com/playlist?list=PL9f9bE-E5OpqK5KRrQNzKtd2tj3DpAyIE&si=AM8JXT38fGnA0Baq, still under development on October 23, 2024.

The Lecturers

Personal Data

Full Name: Loriano Storchi
Nationality: Italian
Institution: University G. D'Annunzio of Chieti-Pescara
E-Email: loriano@storchi.org

Education

I completed my degree in Chemistry with full marks and honors (110/110 cum laude), in November 2000 at the University of Perugia. Soon after my graduation, my first work experience was at the Department of Chemistry at the University of Perugia and ISTM (CNR) from 2000 to 2003. Straight after this first research period, I started my PhD in Chemistry at the University of Perugia which I successfully completed in 3 years and discussed the viva titled "Innovative computational strategies for ab-initio Quantum Chemistry: Grid Computing and novel Green's function techniques"

Professional Experience

I am now Associate Professor at the University G., D'Annunzio of Chieti-Pescara. His research interests are mainly focused in programming and programming languages, HPC (High Performance Computing), Parallel and distributed computing, Chemoinformatics and Machine Learning techniques , Relativistic Density Functional Theory, Over the years he collaborated as programmers with several public and private institutions:

- Molecular Discovery Ltd., UK, developing many commercial programs (a review of which can be found at http://www.moldiscovery.com/)
- CNR (ISTM) The Institute of Molecular Science and Technologies Implementation and optimization of non-relativistic and relativistic DFT codes using parallel computer architectures
- I.N.F.N. (National Institute of Nuclear Physics)- section of Perugia- for evaluating the computational needs for the Einstein Telescope project using the Many-Core (GPU) environment. During this period I developed and optimized codes for gravitational signal analysis within the Einstein Telescope project.
- As a researcher associated with INFN he has been registered as a Software Engineer at CERN as part of the group CMS (Compact Muon Solenoid) of Perugia. In this contest I actively engaged in the development and testing of algorithms track fitting using Associative Memory and FPGA (Field - Programmable Gate Array).
- CRC (Center for Research on Climate and Climate Change) on the Optimization and porting of code meteorological platforms with many cores.

Scientific Activity BIO

My research activity is mainly devoted to computer simulation in both chemical and physical science. In this way I have acquired wide competences in several programming languages and computational and numerical methods, HPC High Performance Computing, parallel and distributed programming, computational architecture (e.g. modern CPU and Memory architectures) , networks and network protocols. Most of my research activity has been focused in Linear Algebra for Large Matrices, Drug Design and chemoinformatics, Machine Learning, Relativistic DFT.

Scientific Publications

Citations	2996
h-index	23
i10-index	49

Full list of publications:

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https://scholar.google.it/citations?user=XtqP4xMAAAAJ&hl=en&oi=ao

Personal Data

Full Name: Simon Cross
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Education
PhD (Chemistry)
University of Nottingham, 1997-2001
Bsc (Hons) Biochemistry & Biological Chemistry
University of Nottingham, 1994-1997

Professional Experience
Snr Scientist & Product Manager. **Molecular Discovery Ltd - Jan 2008-present**
Product Manager. **Tripos, Inc – Jan 2006-Nov 2007**
Application Scientist, UK. **Tripos UK Ltd – Jul 2001-Jan 2006**

Scientific Activity BIO
23 years of experience in the Pharmaceutical Industry with Drug Discovery Software. Co-developer of the software FLAP, including FLAPdock, FLAPpharm and WaterFLAP. Developer of FragExplorer. Co-developer of DeepGRID. Hit-to-lead computational support for EU projects ENABLE and OPTObacteria; consulting on various other projects. Product Manager of FLAP with extensive commercial experience including Acting Commercial Director. As part of a team successfully obtained venture capital for a new start-up.

Scientific Publications
25 publications including 1 book chapter
h-index: 16
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Personal Data

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Education

Master in physics	Ecole Normale Supérieure	1998
PhD in theoretical chemistry	Ecole Normale Supérieure	2001

Professional Experience

Period	Position/Institution/Country
2014-	Director of research/CNRS/France
2014-	Professor of theoretical chemistry/ENS/France
2002-2014	Chargé de recherche/CNRS/France
2001-2002	Postdoctoral researcher/ETH/Switzerland

Scientific Activity BIO

Damien Laage's research area includes chemical reactivity, hydrogen-bond dynamics and spectroscopy in solutions, at interfaces and in biochemical environments. Recently, his interests have focused on understanding chemical reactivity at interfaces using neural network potentials.

Selected Scientific Publications

1. Laage, D.; Elsaesser, T.; Hynes, J.T. "Water dynamics in the hydration shells of biomolecules", *Chem. Rev.* **2017**, *117*, 10694-10725. DOI: 10.1021/acs.chemrev.6b00765
2. de la Puente, M. ; David, R. ; Gomez, A. ; Laage, D. "Acids at the edge: why nitric and formic acid dissociations at air-water interfaces depend on depth and on interface specific area", *J. Am. Chem. Soc.* **2022**, *144*, 10524-10529. DOI: 10.1021/jacs.2c03099
3. de la Puente, M.; Laage, D. How the Acidity of Water Droplets and Films Is Controlled by the Air–Water Interface. *J Am Chem Soc* **2023**, *145*, 25186-25194. DOI: 10.1021/jacs.3c07506
4. David, R.; Tuñón, I.; Laage, D. Competing reaction mechanisms of peptide bond formation in water revealed by deep potential molecular dynamics and path sampling. *J Am Chem Soc* **2024**, *146*, 14213-14224. DOI: 10.1021/jacs.4c03445
5. Gomez, A.; Thompson, W. H.; Laage, D. Neural-network-based molecular dynamics simulations reveal that proton transport in water is doubly gated by sequential hydrogen-bond exchange. *Nature Chem* **2024**, in press. DOI: 0.1038/s41557-024-01593-y

Personal Data

Full Name: Marco Pezzella

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Education

2003-2008: High school degree (Liceo Scienitifico Filippo Masci, Chieti, Italy)

2008-2012: Bachelor degree in "Scienze e Tecnologie Chimiche e dei Materiali"(Università degli Studi dell'Aquila, L'Aquila, Italy).

2010-2011: Erasmus Placement internship (CSIC Cantoblanco, Madrid, Spain)

2012-2014: Master Degree in "Scienze Chimiche" (Università La Sapienza, Roma, Roma, Italy).

2015-2020: PhD in Chemistry (University of Basel, Basel, Switzerland)

Professional Experience

My research experience is grounded in astrochemistry, using both molecular dynamics and spectroscopy models. I developed tools in Fortran, C, R, Python, and Bash. Currently, we are developing a new server using Docker to develop the prototype of the new scientific cluster in Perugia. I presented my findings at various conferences (MUST Switzerland, HRMS, ISGC, *etc*...).

Scientific Activity BIO

Where you find small molecules, atmospheres, and photons, you will find me at my workstation! Since the beginning of my scientific career, I have focused on characterising the potential energy surfaces of large molecules using time-dependent density functional theory (TD-DFT) calculations, evaluating the photodissociation cross sections of small molecules through multi-reference methodologies coupled with the nuclear Schrödinger equation of motion, and conducting molecular dynamics to compare the reactivity of ground states and excited molecules on surfaces in the interstellar medium. As you understood by now, I am interested on how molecules interact in extreme environments.

I have utilised both existing toolkits and developed my own codes. I have employed multiple electronic structure calculation codes (Gaussian, Molpro, etc.), used molecular mechanics codes, primarily CHARMM, and developed modules in the programme suite to reproduce accurate spectroscopy models through kernel ridge regression. I contributed to the development of methodological toolkits to produce reliable photodissociation cross sections of small molecules.

Currently, I am improving my understanding of computer architectures, exploring how cloud computing can improve computational chemistry life, either main focus on how these technologies can be used to produce and analyse data. I am also practicing my quantum mechanics knowledge using the quantum computer available in Perugia.

Scientific Publications:

Pezzella, Marco; Unke, Oliver T; Meuwly, Markus; *J. Phys. Chem. Lett.,* **9,** 8 1822-1826 (2018).

Pezzella, Marco; Meuwly, Markus, *Phys. Chem. Chem. Phys.,* **21**, 11, 6247-6255 (2019).

Ciavardini, Alessandra; Coreno, Marcello; Callegari, Carlo; Spezzani, Carlo; De Ninno, Giovanni; Ressel, Barbara; Grazioli, Cesare; de Simone, Monica; Kivimaki, Antti; Miotti, Paolo; *J. Phys. Chem. A,* **123**, 7,1295-1302 (2019).

Pezzella, Marco; Koner, Debasish; Meuwly, Markus; *J. Phys. Chem. Lett.* , **11**, 6, 2171-2176 (2020).

Pezzella, Marco; El Hage, Krystel; Niesen, Michiel JM; Shin, Sucheol; Willard, Adam P; Meuwly, Markus; Karplus, Martin; *J. Phys. Chem. B* , **124**, 30, 6540-6554 (2020).

Devereux, Mike; Pezzella, Marco; Raghunathan, Shampa; Meuwly, Markus; *J. Chem. Theory Comput.*, **16**, 12, 7267-7280 (2020).

Upadhyay, Meenu; Pezzella, Marco; Meuwly, Markus; *J. Phys. Chem. Lett.,* **12**, 29, 6781-6787 (2021)

Pezzella, Marco; Yurchenko, Sergei N; Tennyson, Jonathan; *Phys. Chem. Chem. Phys*, **23**, 30, 16390-16400 (2021).

Pezzella, Marco; Tennyson, Jonathan; Yurchenko, Sergei N; *Mon. Not. R. Astron. Soc.,* **514**, 3, 4413-4425 (2022).

Salehi, Seyedeh Maryam; Pezzella, Marco; Willard, Adam; Meuwly, Markus; Karplus, Martin; *J. Chem. Phys.,* **158**, 2 (2023).

Bowesman, Charles A; Mizus, Irina I; Zobov, Nikolay F; Polyansky, Oleg L; Sarka, János; Poirier, Bill; Pezzella, Marco; Yurchenko, Sergei N; Tennyson, Jonathan; *Mon. Not. R. Astron. Soc.*, **519**, 4, 6333-6348 (2023).

Tennyson, Jonathan; Pezzella, Marco; Zhang, Jingxin; Yurchenko, Sergei N; *RASTI*, **2**, 1, 231-237 (2023).

Tennyson, Jonathan; Yurchenko, Sergei N; Zhang, Jingxin; Bowesman, Charles A; Brady, Ryan P; Buldyreva, Jeanna; Chubb, Katy L; Gamache, Robert R; Gorman, Maire N; Guest, Elizabeth R; *J. Quant. Spectrosc. Radiat. Transf.*, **326**, 109083 (2024).

Yurchenko, Sergei N; Bowesman, Charles A; Brady, Ryan P; Guest, Elizabeth R; Kefala, Kyriaki; Mitev, Georgi B; Owens, Alec; Perri, Armando N; Pezzella, Marco; Smola, Oleksiy; *Mon. Not. R. Astron. Soc.,* **533**, 3, 3442-3456 (2024).

Personal Data

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Education

- 2005-2009: PhD in physics at the University of Toulouse (France)
- 2000-2005: Bachelor and Master student at the University of Toulouse (France)

Professional Experience

- Since 2011: Assistant Professor at the University of Toulouse (France), LAAS-CNRS
- 2009-2011: Postdoctoral researcher at the University of Stanford (USA) with Pr. Michael Levitt

Scientific Activity BIO

My general research interests include computational structural biology and its application to SAR study, protein/DNA motion and accommodation mechanisms. Specifically, my work is currently focused on the development of new computational methodologies to promote innovative in silico experiments and their application to i) structural flexibility of biomolecules and induced-fit mechanisms; ii) predictive physical models for bio-hybrid materials. I also use classical and hybrid QMMM molecular dynamics to elucidate enzymatic mechanisms, as well as the role of mutations in therapeutic targets (oncogenic mutations, therapeutic resistance, etc).

Scientific Publications

- "*Catalytic Cycle of Type II 4'-Phosphopantetheinyl Transferases*" S Gavalda, A Faille, S Fioccola, MC Nguyen, et al. ACS Catalysis *14*, 8561-8575 (2024).
- "*Unraveling the cGAS catalytic mechanism upon DNA activation through molecular dynamics simulations"* J Soler, P Paiva, MJ Ramos, PA Fernandes, M Brut PCCP 23 (15), 9524-9531 (2021)
- "*Hybrid QM/MM vs Pure MM Molecular Dynamics for Evaluating Water Distribution within p21N-ras and the Resulting GTP Electronic Density"* RH Tichauer, G Favre, S Cabantous, M Brut J Phys Chem B 123 (18), 3935-3944 (2019)

Personal Data

Full Name: HEMERYCK Anne
Nationality: French
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Scientific Activity BIO

Anne Hemeryck is a CNRS research director at LAAS-CNRS in Toulouse in France since 2011, specializing in cutting-edge numerical simulations at the atomic scale for integrated materials in functional devices. Since 2021, she leads the MicroNanoBioTechnology (MNBT) department. Her research focuses on gaining a deep understanding of surfaces and interfaces in nanostructured materials, with particular attention to atomic diffusion and chemical reactions. Anne Hemeryck is expert in the multi-level modeling approach dealin with the complexities and navigating with the limitations of DFT computational codes, molecular dynamics simulations, and the Activation-Relaxation Technique (ART). Her expertise extends to bridging multiple scales, utilizing DFT data to develop macroscale models, and applying methodologies such as kinetic Monte Carlo (kMC), kinetic rate theory, and thermodynamics.

Personal Data

Full Name: Rino Ragno
Nationality: Italian
Institution: Sapienza University of Rome
E-Email: rino.ragno@uniroma1.it

Education
Full professor of medicinal chemistry

Scientific Activity BIO

Rino Ragno is a full professor at the Pharmacy and Medicine Faculty, Sapienza University of Rome where is in charge to teach medicinal chemistry at bachelor, master and PhD level.
In 1989 he finished the 5-year thesis in Pharmaceutical Chemistry and Technology at Sapienza University of Rome. Since 1990 he is enrolled in Sapienza University of Rome. In 1996-1998 he spent two years at the Center for Molecular Design lab directed by prof Garland Marshal ad Washington University of St. Louis (MO - USA). Since 1999, he started a new med chem design lab (www.rcmd.it) with the aim to apply computational methodologies in the field of medicinal chemistry. In 2005 he was awarded by the Italian Chemical Society - Italian division of medicinal chemistry for its research in medicinal chemistry.

From 2000 to 2010 he was an assisant professorand form 2010 20 2022 associate professor at Sapienza University of Rome.
Since 2015 he is in charge of leading the Sapienza MOOC Team to produce new MOOCs from Sapienza.

He published 162 international papers (156 listed in scopus.com) and presented more than 160 posters/oral communications at international meetings.
The main fields of the research are:

- Computational Medicinal Chemistry;
- Development of new QSAR and 3-D QSAR methods;
- Extraction, analysis and biological activity of essential oils;
- Development of scientific web sites (www.3d-qsar.com, db.3d-qsar.com and www.ai4essoil.com).
- Drug Design through computational techniques: machine learning, molecular docking
- Virtual Screening to select new potential bioactive compounds
- Application of machine learning algorithms for the design of new bioactive molecules
- Application of machine learning algorithms for analysis of complex mixtures and correlation with their biological activity

Personal Data

Full Name: Ferran Feixas
Nationality: Spain
Institution: University of Girona, Spain
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Education

Degree in Chemistry	Universitat de Girona	2005
Master in Theoretical and Computational Chemistry	Universitat de Girona	2007
PhD in Chemistry	Universitat de Girona	2011

Professional Experience

Period	Position/Institution/Country/Interruption cause
2006-2011	PhD fellowship, Universitat de Girona, Spain
2012-2014	Beatriu de Pinós Postdoctoral Researcher, University of California San Diego, United States
2014-2015	Beatriu de Pinós Postdoctoral Researcher, Universitat de Girona, Spain
2015-2017	Marie Curie Individual Fellowship, Universitat de Girona, Spain
2018-2019	Senior Researcher, Universitat de Girona, Spain
2019-2022	JIN Researcher, UdG, Spain
2022 – now	Ramón y Cajal Researchers, UdG, Spain

Scientific Activity BIO

Dr. Ferran Feixas is currently a Ramón y Cajal researcher at the Institute of Computational Chemistry and Catalysis (IQCC) at the University of Girona.

Dr. Feixas obtained the PhD in Chemistry in 2011 at the University of Girona (Spain) under the supervision of Prof. Miquel Solà, Prof. Jordi Poater, and Dr. Eduard Matito. During the PhD thesis (FPU fellowship), he worked on the application and development of computational chemistry tools for the analysis of chemical bonding and aromaticity. In this period, he carried out research stays in France (Prof. B. Silvi) and Germany (Prof. L. González). Upon graduation, Ferran joined the University of California, San Diego (UCSD, USA) for a two-year postdoctoral position with a Beatriu de Pinós (BP) fellowship with Prof. J. Andrew McCammon group. At UCSD, he gained expertise in computational biochemistry, drug discovery, and biomolecular simulations. In particular, he worked on the incorporation of molecular dynamics simulations into computer-aided drug discovery protocols and the development and application of enhanced sampling techniques based on accelerated molecular dynamics (aMD) simulations for studying protein folding. In 2015, he joined the IQCC (UdG) with a Marie Curie Individual Fellowship for studying the molecular basis of (bio)molecular recognition and self-assembly with MD simulations. In 2018, he became a Senior Researcher and co-PI of the CompBioLab research group together with Prof. Sílvia Osuna (ICREA/UdG) working in the application of enhanced sampling techniques for exploring the role of conformational dynamics in enzyme catalysis and improving computational protocols for enzyme design.

In 2019, Ferran was awarded with a Retos de la Sociedad JIN project and in 2021 a Ramón y Cajal fellowship. The goal of the research group is to develop novel strategies combining different simulation techniques that account for the dynamics of (bio)molecules to reconstruct step by step (bio)chemical and (bio)molecular recognition processes and, then, harness this information to enhance the discovery and rational design of novel drugs, supramolecular complexes, and/or (bio)catalysts. Currently the research team is composed by 6 PhDs and 1 postdoc. He has supervised a total of 15 undergraduate, 4 master, and 2 PhD students. He also established collaboration with Fundació Jaume Vicens Vives in the framework of the CIMs-Cellex program to disseminate general chemistry to International Baccalaureate high school students at Institut Jaume Vicens Vives. Finally, Ferran was one of the organizers of the conference Girona Seminar 2022 in Biocatalysis and serves as reviewer for several journals, funding agencies, and RES.

Scientific Publications

1. Calvó-Tusell, C.; Maria-Solano, M. A.; Osuna, S.; Feixas, F.; "Time Evolution of the Millisecond Allosteric Activation of Imidazole Glycerol Phosphate Synthase" *J. Am. Chem. Soc.* **2022**, 144, 7146-7159

2. Girame, H.; Garcia-Borràs, M; Feixas, F.; "Changes in Protonation States of In-Pathway Residues can Alter Ligand Binding Pathways Obtained From Spontaneous Binding Molecular Dynamics Simulations" *Front. Mol. Biosci.* **2022**, 9, 922361

3. López-Coll, R.; Álvarez-Yebra, R.; Feixas, F.; Lledó, A. "Comprehensive Characterization of the Self-Folding Cavitand Dynamics" *Chem.- Eur. J.* **2021**, 27, 10099-10106

4. Codony, S.; Calvó-Tusell, C.; Valverde, E.; …; Feixas, F.; Vazquez, S. "From the Design to the In Vivo Evaluation of Benzohomoadamantane-Derived Soluble Epoxide Hydrolase Inhibitors for the Treatment of Acute Pancreatitis" *J. Med. Chem.* **2021**, 64, 5429-5446

5. García-Simón, C.; Colomban, C.; Çetin, Y.‡; …; Jiménez-Barbero, J.; Feixas, F.; Ribas, X. "Complete dynamic reconstruction of the C60, C70, and (C59N)2 encapsulation into an adaptable supramolecular nanocapsule" *J. Am. Chem. Soc.* **2020**, 142, 16051-16063

6. Tomás-Loba, A.; Manieri, E.; González-Terán, B.; …; Sabio, G. "p38gamma is essential for cell cycle progression and liver tumourigenesis" *Nature* **2019**, 568, 557-560

7. Curado-Carballada, C.; Feixas, F.; Iglesias-Fernández, J.; Osuna, S.* "Hidden conformations in Aspergillus niger Monoamine Oxidase key for catalytic efficiency" *Angew. Chem. Int. Ed.* **2019**, 58, 3097-3101

8. Arqué, X.; Romero-Rivera, A.; Feixas, F.; Patiño, T.; Osuna, S., Sánchez, S. "Intrinsic enzymatic properties modulate the self-propulsion of micromotors" *Nat. Commun.* **2019**, 10, 2826

9. Miao, Y.; Feixas, F.‡; Eun, C.; McCammon, J.A.; "Accelerated Molecular Dynamics Simulations of Protein Folding", *J. Comp. Chem*. **2015**, 36, 1536-1549

Personal data

Full Name: Ioana ILIE
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Scientific Activity BIO

Dr. Ioana M. ILIE is as an Assistant Professor within the Computational Chemistry group at the University of Amsterdam. Having obtained her PhD in Computational Biophysics from the University of Twente, she furthered her academic pursuits as a postdoctoral researcher at the University of Zurich, focusing on the intricate biochemical aspects of proteomics. The research in her group is centered around multiscale simulations of biomolecular systems. Specifically, her team employs rational design and multiscale simulations to achieve three main objectives: (1) understanding and controlling the aggregation mechanisms of polypeptides and their interactions within the biological environment, (2) designing peptide-based therapeutics for neurodegenerative diseases, and (3) developing intelligent (bio)materials with responsive properties.

Personal data

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- since 2016: Head of the "General Chemistry" learning unit@UT3, 600 students, 20 researcher-lecturers
- since 2018: Head of the Nanoscale Science and Engineering Graduate School of Research
- since 2011: Member of the executive committee of the NanoX project (formerly NEXT Labex)
- since 2006: Head of the physical and chemical modelling group of the LPCNO (6 tenured researchers and ~5-10 PhD students and post-doctoral fellows)
- since 2003: Full Professor of Theoretical and Physical Chemistry at the Paul Sabatier University, Toulouse, France
- 2002: "Accreditation to supervise research" (HDR), carried out at the Laboratoire de Physique Quantique (IRSAMC-UPS), November 2002, 25th. "Chemical functional groups: actors or spectators?", supervisor: Dr. J.-P. Daudey
- 1993-2003: Assistant Professor at the Paul Sabatier University, Toulouse, France
- 1997-1999: Two-year full-time research position at the French National Centre for Scientific Research (Centre national de la recherche scientifique, CNRS)
- 1990-1993: PhD in Theoretical Chemistry, carried out at the Laboratoire de Physique Quantique (IRSAMC, Toulouse, France), supervisor: Fernand Spiegelmann, "Structure, Thermodynamics and Spectroscopy of Small Sodium Clusters" - Adjunct lecturer

Scientific Activity BIO

Romuald Poteau is **Professor of Computational and Physical Chemistry** at the Toulouse 3 - Paul Sabatier University, France. He received a PhD in theoretical chemistry from the same university in 1993, followed by his professorial thesis in 2002, titled "Functional chemical groups: actors or spectators?". His research interests include **Metal clusters and Nanoclusters, Coordination Chemistry and Theoretical Inorganic Chemistry, Global Optimization Methods, Spectroscopic Properties of Molecules**, mainly using **DFT** methods and other computational methods. Since a decade or so, he attempted to bridging the gap between molecular inorganic chemistry and nanochemistry, with a focus on **nanocatalysis**. He recently developed a new research activity on **machine learning applied to computational chemistry**. On the educational side, he is involved in undergraduate and graduate chemistry and physical chemistry courses. Since a couple of years, he also develops two freely available web resources that may be used as teaching tools: **vChem3D**, a website that shows **3D representations of molecules and associated properties** (http://www.vchem3d.com); **pyPhysChem**, a github repository focusing on using **Python** for physical chemistry, with a range of tutorials and practical examples provided in Jupyter Notebooks and that covers various topics such as **molecular modeling, data analysis, and machine learning** applied to chemistry (https://github.com/rpoteau/pyPhysChem). He is the **director of the Nanoscale Science and Engineering Graduate School of Research in Toulouse**, and the director of studies of the Nanoscale Science and Engineering Master of Science.

Representative Scientific Publications

- R. Poteau, I. Ortega, F. Alary, A. R. Solis, J.-C. Barthelat, J.-P. Daudey, Effective Group Potentials. 1. Method, *J. Phys. Chem. A* **2001**, *105*, 198-205, (https://dx.doi.org/10.1021/jp002500k).
- L. Cusinato, L. M. Martinez-Prieto, B. Chaudret, I. del Rosal, R. Poteau, Theoretical characterization of the surface composition of ruthenium nanoparticles in equilibrium with syngas, *Nanoscale* **2016**, *8*, 10974-10992, (https://dx.doi.org/10.1039/C6NR01191H).
- R. González-Gómez, L. Cusinato, C. Bijani, Y. Coppel, P. Lecante, C. Amiens, I. del Rosal, K. Philippot, R. Poteau, Carboxylic acid-capped ruthenium nanoparticles: experimental and theoretical case study with ethanoic acid, *Nanoscale* **2019**, *11*, 9392-9409, (https://dx.doi.org/10.1039/c9nr00391f).

Ramamoorthy, R. K.; Soulantica, K.; Del Rosal, I.; Arenal, R.; Decorse, P.; Piquemal, J.-Y.; Chaudret, B.; Poteau, R.; Viau, G. Ruthenium Icosahedra and Ultrathin Platelets: The Role of Surface Chemistry on the Nanoparticle Structure. *Chem. Mater.* **2022**, *34* (7), 2931–2944 (https://doi.org/10.1021/acs.chemmater.1c03452).